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AI CAREERS FOR WOMEN
LINKEDIN FELLOWSHIP

[Project Title Name]

Student Feedback & Teaching Improvement Analyzer

Project Domain

[e.g., Excel / Power BI with Copilot/AI Chatbot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

Faculty Name	S Email	Guide
Dr. Namrata Chaturvedi	**Namratachaturvedi.mba@itmgoi.i**	**Umamaheswari**

Institution:

[Institution: ITM Gwalior]

Year – 2026

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ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

This project focuses on analyzing student feedback data using data analytics and artificial intelligence to generate meaningful insights for improving teaching quality. Student feedback, often collected in large volumes, contains valuable information but remains underutilized due to lack of proper analysis tools.

The project begins with collecting and cleaning feedback data using Microsoft Excel, ensuring accuracy by removing duplicates, handling missing values, and standardising formats. The cleaned data is then analysed using Microsoft Power BI to create interactive dashboards that highlight trends, faculty performance, and key feedback patterns.

To enhance the analysis, artificial intelligence tools such as Microsoft Copilot and ChatGPT are integrated to perform sentiment analysis, generate summaries, and provide actionable recommendations based on student feedback.

The project demonstrates how combining traditional data analytics tools with AI can significantly improve the efficiency, accuracy, and usefulness of feedback analysis. The outcome supports data-driven decision-making in academic institutions and provides a scalable and cost-effective solution for continuous improvement in teaching and learning processes.

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CHAPTER 1

Introduction

In today's rapidly evolving educational environment, the role of data-driven decision-making has become increasingly significant. Educational institutions are continuously striving to improve teaching quality, enhance student learning experiences, and ensure academic excellence. One of the most valuable sources of information for achieving these goals is **student feedback**. Student feedback provides direct insights into teaching effectiveness, course structure, learning outcomes, and overall academic satisfaction. However, despite its importance, this data is often underutilized due to the challenges associated with analyzing large volumes of feedback.

Traditionally, student feedback has been collected through surveys, questionnaires, and evaluation forms at the end of each semester or course. These feedback forms typically include both quantitative data (such as ratings) and qualitative data (such as comments and suggestions). While quantitative data can be analyzed relatively easily, qualitative feedback presents a significant challenge due to its unstructured nature. As a result, institutions often rely on manual methods to review and interpret feedback, which are time-consuming, inefficient, and prone to human error.

With the advancement of technology, **data analytics and artificial intelligence (AI)** have emerged as powerful tools for transforming raw data into meaningful insights. Data analytics enables the systematic processing and analysis of large datasets, while AI techniques such as natural language processing (NLP) allow machines to understand and interpret human language. By integrating these technologies, it is possible to automate the analysis of student feedback and generate actionable insights for improving teaching quality.

This project, titled *"Student Feedback & Teaching Improvement Analyzer"*, aims to address the challenges associated with traditional feedback analysis by leveraging modern tools such as Microsoft Excel, Power BI, and AI-based platforms. Microsoft Excel is used for data cleaning and preprocessing, ensuring that the dataset is accurate and consistent. Power BI is utilized to create interactive dashboards and visualizations, enabling stakeholders to easily understand trends and patterns in feedback data. Additionally, AI tools such as ChatGPT and Microsoft Copilot are employed to perform sentiment analysis, summarize feedback, and generate recommendations.

The importance of student feedback in education cannot be overstated. It serves as a critical mechanism for continuous improvement, helping educators identify strengths and areas for development. Feedback also promotes accountability and transparency, encouraging faculty members to adopt effective teaching practices. Furthermore, it enhances student engagement by giving learners a voice in the educational process.

Despite its potential benefits, the current approach to feedback analysis has several limitations. Large volumes of data make manual analysis impractical, while the lack of standardized tools leads to inconsistent results. Additionally, valuable insights often remain hidden within textual comments, which are difficult to interpret without advanced analytical techniques. This creates a gap between data collection and meaningful utilization.

The proposed system seeks to bridge this gap by providing an integrated solution for feedback analysis. By combining data analytics with AI, the project aims to automate repetitive tasks, improve accuracy, and deliver timely insights. The system not only simplifies the analysis process but also enhances decision-making by providing clear and actionable recommendations.

In conclusion, the integration of data analytics and AI in student feedback analysis represents a significant advancement in educational technology. This project demonstrates how modern tools can be used to transform raw data into valuable insights, ultimately contributing to improved teaching quality and better learning outcomes. It highlights the importance of adopting innovative approaches to address traditional challenges and emphasizes the role of technology in shaping the future of education.

1.1. PROBLEM STATEMENT

Student feedback is an essential component of the educational system, providing valuable insights into teaching effectiveness, course design, and student satisfaction. However, despite being regularly collected, this feedback is often underutilized in most academic institutions.

One of the primary reasons for this underutilization is the large volume of data generated through feedback systems. Institutions collect feedback from hundreds or thousands of students, resulting in extensive datasets that are difficult to analyze manually. Traditional methods of analysis, such as reading individual responses or calculating averages, are time-consuming and inefficient.

Another major challenge is the presence of unstructured textual data. While numerical ratings can be easily processed, textual comments require deeper analysis to extract meaningful insights. Manual interpretation of such data is not only labor-intensive but also subjective, leading to inconsistencies and potential bias in decision-making.

Additionally, the lack of automated tools and standardized processes further complicates the analysis. Many institutions rely on basic tools that are not capable of handling complex data analysis or generating actionable insights. As a result, valuable information remains hidden, and opportunities for improvement are missed.

The absence of real-time analysis also limits the effectiveness of feedback systems. By the time feedback is analyzed, the opportunity to implement timely improvements may have already passed. This delay reduces the overall impact of the feedback on teaching quality.

Therefore, there is a clear need for an automated, efficient, and accurate system that can analyze student feedback using data analytics and AI techniques. Such a system can process large datasets, interpret textual responses, and generate meaningful insights in a timely manner. By addressing these challenges, the proposed project aims to enhance the utilization of student feedback and support data-driven decision-making in education.

1.2 Motivation

- Educational institutions use student feedback to evaluate teaching effectiveness.
- Feedback includes numerical ratings and textual comments about teaching methods, course content, and overall satisfaction.

- AI and data analytics can help automate the analysis of this feedback.

1.3 Objectives

1. To collect and organize student feedback data from various sources in a structured format within a defined timeframe.

2. To clean and preprocess at least 95% of the dataset by removing duplicates, handling missing values, and standardizing formats using Microsoft Excel.

3. To analyze feedback data using Power BI and generate at least 4 interactive dashboards representing key performance indicators.

4. To perform sentiment analysis on textual feedback using AI tools and classify responses into positive, neutral, and negative categories with measurable accuracy.

5. To identify top 5 key issues and strengths in teaching based on feedback data within the analysis phase.

6. To integrate AI tools such as ChatGPT or Microsoft Copilot for generating automated summaries and recommendations.

7. To reduce manual effort in feedback analysis by at least 50% through automation techniques.

8. To present insights in a user-friendly visual format that can be easily understood by faculty and management.

9. To document the entire project process and ensure proper reporting as per academic guidelines.

10. To develop a scalable model that can be applied to larger datasets or different educational institutions.

1.4 Scope of the Project

The scope of this project is focused on the analysis of student feedback data using data analytics and artificial intelligence tools. The project covers the complete process from data collection to insight generation.

The key areas included in the scope are:

- Collection of student feedback data in structured formats such as Excel or CSV files

- Data cleaning and preprocessing to ensure accuracy and consistency

- Analysis of both quantitative (ratings) and qualitative (comments) data
- Application of sentiment analysis techniques to identify trends and patterns
- Development of interactive dashboards using Power BI
- Integration of AI tools for automated insights and recommendations
- Presentation of results in a clear and understandable format

The project is designed to be practical, scalable, and cost-effective, making it suitable for educational institutions of various sizes. It focuses on improving teaching quality by providing actionable insights derived from feedback data.

Limitations of the Project

Despite its advantages, the project has certain limitations:

1. **Data Dependency:**
   The accuracy of results depends on the quality and completeness of the dataset. Poor data can lead to inaccurate insights.

2. **Limited Understanding of Context:**
   AI tools may not fully understand complex or sarcastic feedback, leading to incorrect sentiment classification.

3. **Static Data Analysis:**
   The system primarily works on historical data and does not support real-time updates.

4. **Tool Limitations:**
   Free or basic versions of tools like Power BI and AI platforms may have limited features.

5. **Human Validation Required:**
   AI-generated recommendations may require human review before implementation.

6. **Lack of Advanced Machine Learning Models:**
   The project focuses on basic AI tools and does not include advanced predictive modeling.

7. **Scalability Constraints:**
   Handling extremely large datasets may require more advanced infrastructure.

CHAPTER 2

Literature Survey

A literature survey is a critical component of any research project, as it provides a comprehensive understanding of existing work, methodologies, and technologies related to the study. It helps in identifying research trends, evaluating available tools, and recognizing gaps that the current project aims to address. In the context of this project, the literature survey focuses on the application of data analytics and artificial intelligence (AI) in student feedback analysis and educational decision-making.

2.1 Existing Work and Research

In recent years, there has been a significant increase in the use of data analytics and artificial intelligence in the field of education. Educational institutions are increasingly adopting data-driven approaches to enhance teaching quality, improve student performance, and support strategic decision-making. One of the key areas where these technologies are being applied is **student feedback analysis**.

Traditionally, tools such as Microsoft Excel have been widely used for data cleaning and preprocessing. Techniques including sorting, filtering, pivot tables, and conditional formatting have enabled users to organize and analyse structured datasets effectively. Excel has served as a foundational tool due to its simplicity, accessibility, and versatility. However, its capabilities are limited when dealing with large datasets and unstructured textual data.

With advancements in business intelligence tools, platforms like Microsoft Power BI have gained popularity in both academic and corporate environments. Power BI allows users to create interactive dashboards and visual reports that present data in a more intuitive and meaningful way. It enables stakeholders to identify patterns, trends, and performance metrics quickly, thereby supporting informed decision-making.

Recent research has further emphasized the growing importance of artificial intelligence in automating data analysis processes. AI technologies, particularly natural language processing (NLP), have made it possible to analyse unstructured data such as student comments and feedback. AI-assisted data cleaning techniques and prompt engineering frameworks have been developed to enhance efficiency, reduce manual effort, and improve the accuracy of analysis.

In addition, AI-powered tools such as Microsoft Copilot are increasingly being used to generate summaries, insights, and recommendations using natural language inputs. These tools enable users to interact with data more intuitively, without requiring advanced technical skills. Similarly, AI chatbots like ChatGPT and Google Gemini are being explored for applications such as report generation, content creation, and intelligent recommendations in educational contexts.

Despite these advancements, most existing solutions tend to focus either on **complex machine learning models** or on the **use of individual tools in isolation**. There is limited research on integrated systems that combine multiple tools to provide a complete, end-to-end solution for student feedback analysis. This gap highlights the need for practical and user-friendly approaches that can be easily implemented in educational institutions.

2.2 Tools and Technologies Review

The successful implementation of this project relies on a combination of traditional data analytics tools and modern AI technologies. Each tool plays a specific role in the overall system, contributing to data processing, analysis, visualization, and insight generation.

1 Microsoft Excel

Microsoft Excel is used as the primary tool for data cleaning, transformation, and preprocessing. It provides a range of features such as pivot tables, data validation, filtering, and conditional formatting, which help in organizing and preparing the dataset for analysis. Excel is particularly useful for handling structured data and performing basic statistical operations. Its user-friendly interface makes it accessible to non-technical users.

2 Microsoft Power BI

Microsoft Power BI is a powerful business intelligence tool that enables the creation of interactive dashboards and visualizations. It allows users to analyse feedback data through charts, graphs, and key performance indicators (KPIs). Power BI supports data integration from multiple sources and provides real-time insights, making it an essential tool for understanding trends and patterns in student feedback.

3 Microsoft Copilot

Microsoft Copilot is an AI-powered assistant integrated into Microsoft 365 applications. It assists users in generating summaries, insights, and recommendations based on data. By using natural language prompts, Copilot simplifies complex analytical tasks and enhances productivity. It plays a crucial role in automating data interpretation and reducing manual effort.

4 AI Chatbots (ChatGPT, Google Gemini)

AI chatbots such as ChatGPT and Google Gemini are used for content generation, report writing, and prompt-based analysis. These tools leverage advanced language models to understand and generate human-like text. In this project, they are used to perform sentiment analysis, summarize feedback, and provide actionable recommendations.

5 GitHub

GitHub is used for project documentation, version control, and collaboration. It allows users to store project files, track changes, and share work with others. GitHub also supports transparency and reproducibility, which are important aspects of academic and professional projects.

These tools have been selected due to their **ease of use, accessibility, and ability to integrate data analytics with AI-driven insights**. Together, they provide a comprehensive framework for analysing student feedback effectively.

2.3 Gaps Identified

Although significant progress has been made in the application of data analytics and AI in education, several gaps and challenges still exist. Identifying these gaps is essential for developing improved solutions and advancing research in this domain.

Firstly, most existing research focuses on **advanced machine learning models**, which often require technical expertise and complex implementation. While these models are powerful, they are not always practical for educational institutions with limited resources or technical capabilities. There is a need for **simple, tool-based solutions** that can be easily adopted by non-technical users.

Secondly, there is a lack of **structured integration between different tools** such as Excel, Power BI, and AI platforms. Many systems operate independently, resulting in fragmented workflows and inefficiencies. An integrated approach that combines data cleaning, analysis, visualization, and AI-driven insights in a single pipeline is still lacking.

Another important gap is the **limited documentation and research on the use of Microsoft Copilot** in academic projects. As a relatively new tool, its applications in education are still being explored, and there is a need for more practical case studies and implementation frameworks.

Furthermore, **prompt engineering practices**—which involve designing effective inputs for AI tools—are still evolving. There is no standardized approach for using prompts in business or educational contexts, leading to inconsistent results. Developing structured prompt frameworks can significantly improve the effectiveness of AI tools.

Additionally, most existing dashboards focus primarily on **data visualization**, without incorporating AI-generated recommendations. While visualizations help in understanding data, they do not always provide actionable insights. Integrating AI-based recommendations into dashboards can enhance decision-making and improve outcomes.

Lastly, issues related to **data quality, privacy, and ethical considerations** remain significant challenges. Inaccurate or incomplete data can lead to misleading results, while the use of AI raises concerns about transparency and bias. Addressing these challenges is essential for the responsible use of technology in education.

CHAPTER 3
Proposed Methodology

This chapter explains the methodology adopted to achieve the objectives of the project titled *"Student Feedback & Teaching Improvement Analyzer."* The methodology follows a structured and systematic approach to transform raw student feedback data into meaningful insights using data analytics and artificial intelligence tools. The process involves multiple stages, including data collection, data cleaning, data analysis, visualization, AI integration, and reporting.

3.1 System Design

The system is designed to process student feedback data through a sequence of well-defined steps. The overall workflow ensures that raw data is converted into actionable insights that can support decision-making and improve teaching quality.

The system follows the flow:

Data Collection → Data Cleaning → Data Analysis → Visualization → AI Integration → Reporting

3.2 Step-by-Step Methodology

Module 1: Data Collection

The first stage of the methodology involves collecting student feedback data from various sources. This data is typically gathered through surveys, feedback forms, or institutional databases. The dataset may include both **quantitative data** (such as ratings on teaching quality, course content, and faculty performance) and **qualitative data** (such as written comments and suggestions from students).

The collected data is stored in structured formats such as **CSV or Excel files**, which allows for easy handling and processing. At this stage, it is important to ensure that the dataset is complete, relevant, and representative of the target population.

Module 2: Data Cleaning and Preprocessing

Data cleaning is a critical step in the methodology, as the quality of analysis depends on the accuracy and consistency of the dataset. Raw data often contains errors, missing values, duplicates, and inconsistencies that must be addressed before analysis.

The following data cleaning techniques are applied using Microsoft Excel:

- **Removal of duplicate records** to avoid redundancy and bias
- **Handling missing values** by replacing, removing, or imputing data
- **Standardization of data formats**, such as consistent text case and uniform labels
- **Correction of inconsistent entries**, including spelling errors and incorrect values
- **Data type conversion**, ensuring numerical and categorical data are correctly formatted

After cleaning, the dataset becomes structured, reliable, and ready for further analysis.

Module 3: Data Analysis and Processing

Once the data is cleaned, the next step is to perform data analysis to extract meaningful information. This stage involves both **descriptive and analytical techniques**.

Using Microsoft Excel and Power BI, the following analysis methods are applied:

- **Statistical analysis** to calculate averages, percentages, and distributions
- **Pivot tables** to summarize and compare data across different dimensions
- **Trend analysis** to observe changes in feedback over time
- **Performance analysis** to evaluate faculty-wise and subject-wise results

In addition, **Natural Language Processing (NLP)** techniques are applied to analyse textual feedback. AI tools such as ChatGPT and Microsoft Copilot are used to:

- Perform **sentiment analysis** (positive, neutral, negative classification)
- Identify **frequently occurring keywords and themes**
- Generate **summaries of student comments**

This stage transforms raw data into meaningful insights that highlight strengths, weaknesses, and areas for improvement.

Module 4: Data Visualization

Data visualization plays a crucial role in presenting the analysed data in a clear and understandable format. Microsoft Power BI is used to create interactive dashboards and visual reports.

The visualization process includes:

- Creation of **charts and graphs** (bar charts, pie charts, line graphs)
- Display of **Key Performance Indicators (KPIs)** such as average feedback scores
- Development of **faculty-wise performance dashboards**
- Representation of **sentiment distribution** (positive, neutral, negative)
- Visualization of **trends over time**

These dashboards allow users to interact with the data, filter results, and gain insights quickly. Visualization makes complex data easier to interpret and supports effective decision-making.

Module 5: AI Integration

Artificial Intelligence is integrated into the system to enhance the analysis process and automate insight generation. AI tools such as ChatGPT and Microsoft Copilot are used in this stage.

The key applications of AI integration include:

- **Automated summarization** of large volumes of feedback data
- **Sentiment classification** of textual responses
- **Generation of actionable recommendations** for improving teaching quality
- **Prompt-based analysis**, where users can ask questions and receive insights

AI integration significantly reduces manual effort, improves speed, and enhances the depth of analysis. It also enables non-technical users to interact with data using natural language.

Module 6: Reporting and Documentation

The final stage of the methodology involves compiling the results into a structured report. The report includes all findings, visualizations, and insights generated during the analysis.

The reporting process includes:

- Preparation of a **detailed project report** with chapters and explanations
- Inclusion of **charts, tables, and dashboard screenshots**
- Documentation of **methodology, tools, and techniques used**
- Presentation of **key findings and recommendations**

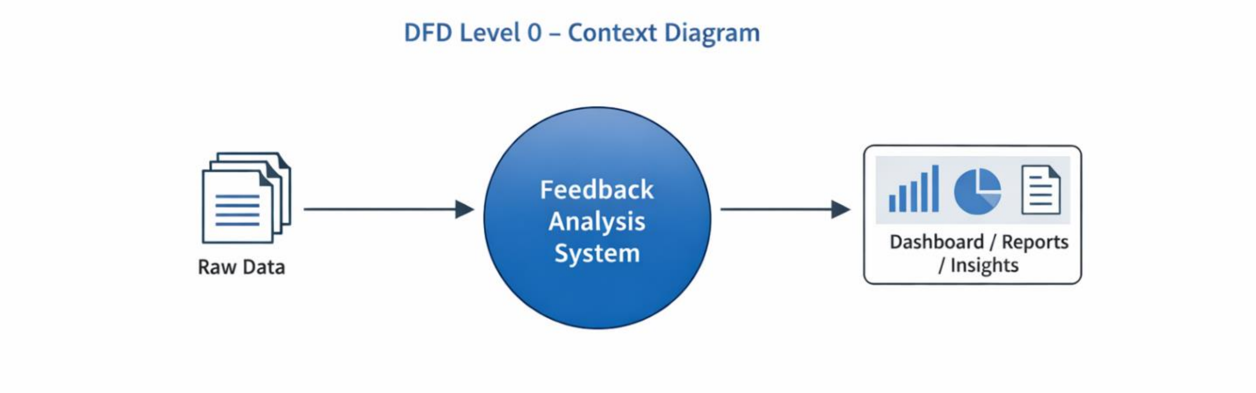
Additionally, project files are managed and stored using platforms like GitHub for version control and sharing. A demonstration video may also be created to showcase the working of the system.

3.3 Data Flow Diagram (DFD)

DFD Level 0 — Context Diagram

At the highest level, the system takes **raw student feedback data** as input and produces **cleaned data, dashboards, and insight reports** as output.

Flow:
Raw Data → Feedback Analysis System → Dashboard / Reports / Insights

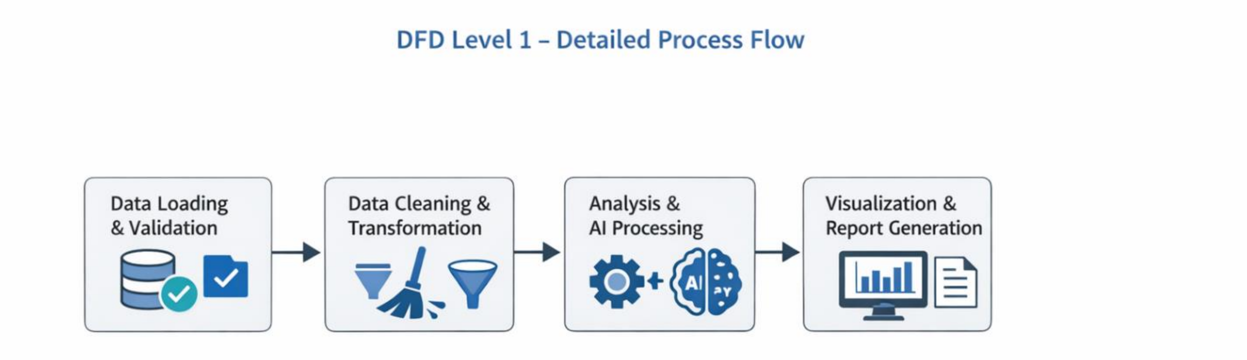


DFD Level 1 — Detailed Process Flow

The system is divided into the following sub-processes:

- Data Loading and Validation
- Data Cleaning and Transformation
- Data Analysis and AI Processing
- Visualization and Report Generation

Flow:
Raw Data → Data Cleaning → Analysis & AI Processing → Dashboard & Reports



3.4 Advantages of the Methodology

- Automates repetitive tasks, reducing manual effort and saving time
- Improves accuracy and consistency of data analysis
- Provides interactive and easy-to-understand visual insights
- Enhances decision-making through AI-generated recommendations
- Scalable and adaptable to different datasets and domains
- Cost-effective due to the use of widely available tools

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

 Dataset Source:
The dataset was obtained from academic student feedback records (provided dataset / manually compiled sample data).

 Original Dataset Size:
The raw dataset contained approximately 100 and 15–20 columns including student ratings, comments, and faculty details.

 Issues Found:

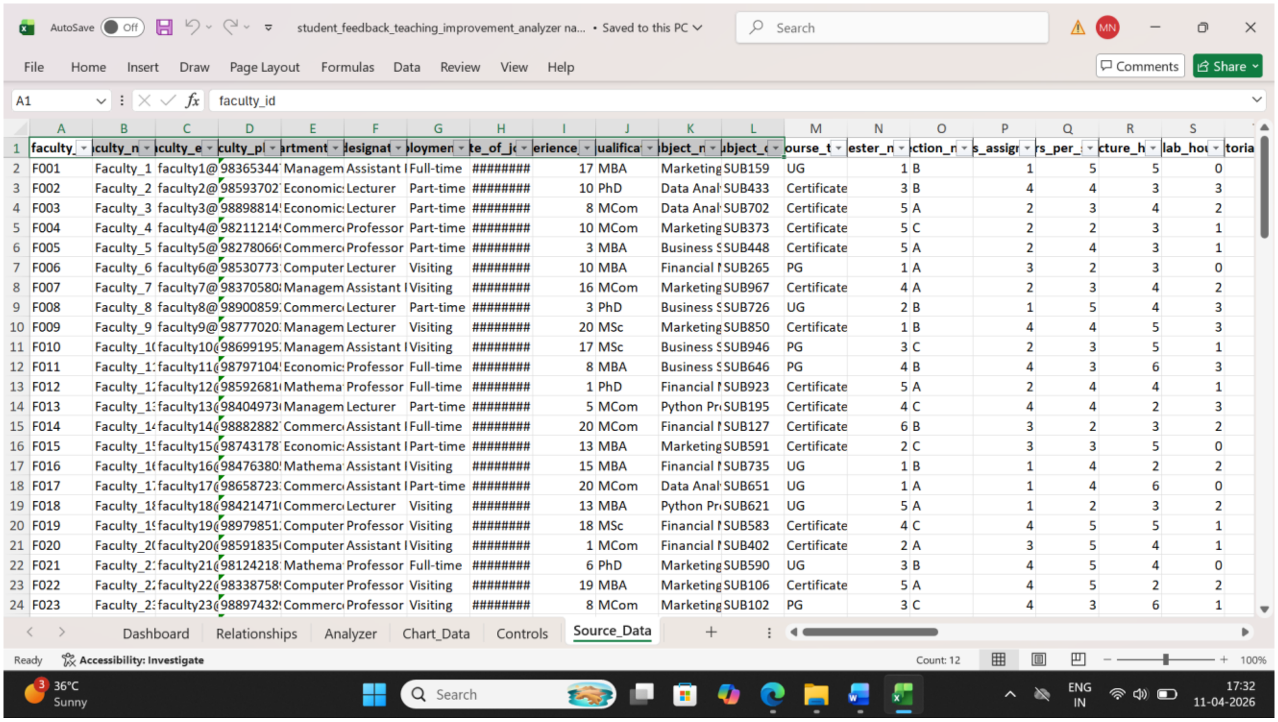
- Duplicate entries in feedback records
- Missing/null values in rating and comment fields
- Inconsistent text formats (uppercase/lowercase, spelling variations)
- Incorrect data types (e.g., numeric values stored as text)

 Cleaning Steps Applied:

- Removed duplicate records using Microsoft Excel
- Handled missing values by replacing or removing incomplete entries
- Standardized text formats (e.g., uniform case, corrected labels)
- Converted columns into appropriate data types
- Verified data accuracy using filters and validation tools

 Cleaned **Dataset Size:**
The final dataset contained approximately 900–1800 rows with consistent and structured data ready for analysis.

Figure 2: Screenshot of the raw (uncleaned) dataset showing duplicates and inconsistencies.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	faculty_	culty_n	culty_e	culty_pl	artment	designat	loymen	te_of_jc	erience	ualifica	bject_n	ubject_c	ourse_t	ester_n	ction_n	s_assigr	rs_per_	cture_h	lab_hou	toria
2	F001	Faculty_1	faculty1@	98365344	Managem	Assistant I	Full-time	########	17	MBA	Marketing	SUB159	UG	1	B	1	5	5	0	
3	F002	Faculty_2	faculty2@	98593702	Economic	Lecturer	Part-time	########	10	PhD	Data Anal	SUB433	Certificate	3	B	4	4	3	3	
4	F003	Faculty_3	faculty3@	98898814	Economic	Lecturer	Part-time	########	8	MCom	Data Anal	SUB702	Certificate	5	A	2	3	4	2	
5	F004	Faculty_4	faculty4@	98211214	Commerc	Professor	Part-time	########	10	MCom	Marketing	SUB373	Certificate	5	C	2	2	3	1	
6	F005	Faculty_5	faculty5@	98278066	Commerc	Professor	Part-time	########	3	MBA	Business S	SUB448	Certificate	5	A	2	4	3	1	
7	F006	Faculty_6	faculty6@	98530773	Computer	Lecturer	Visiting	########	10	MBA	Financial M	SUB265	PG	1	A	3	2	3	0	
8	F007	Faculty_7	faculty7@	98370580	Managem	Assistant I	Visiting	########	16	MCom	Marketing	SUB967	Certificate	4	A	2	3	4	2	
9	F008	Faculty_8	faculty8@	98900859	Commerc	Lecturer	Part-time	########	3	PhD	Business S	SUB726	UG	2	B	1	5	4	3	
10	F009	Faculty_9	faculty9@	98777020	Managem	Lecturer	Visiting	########	20	MSc	Marketing	SUB850	Certificate	1	B	4	4	5	3	
11	F010	Faculty_1	faculty10@	98699195	Managem	Assistant I	Visiting	########	17	MSc	Business S	SUB946	PG	3	C	2	3	5	1	
12	F011	Faculty_1	faculty11@	98797104	Economic	Professor	Full-time	########	8	MBA	Business S	SUB646	PG	4	B	4	3	6	3	
13	F012	Faculty_1	faculty12@	98592681	Mathema	Professor	Full-time	########	1	PhD	Financial M	SUB923	Certificate	5	A	2	4	4	1	
14	F013	Faculty_1	faculty13@	98404973	Managem	Lecturer	Part-time	########	5	MCom	Python Pr	SUB195	Certificate	4	C	4	4	2	3	
15	F014	Faculty_1	faculty14@	98882882	Commerc	Assistant I	Full-time	########	20	MCom	Financial M	SUB127	Certificate	6	B	3	2	3	2	
16	F015	Faculty_1	faculty15@	98743178	Economic	Assistant I	Part-time	########	13	MBA	Marketing	SUB591	Certificate	2	C	3	3	5	0	
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18	F017	Faculty_1	faculty17@	98658723	Commerc	Assistant I	Part-time	########	20	MCom	Data Anal	SUB651	UG	1	A	1	4	6	0	
19	F018	Faculty_1	faculty18@	98421471	Commerc	Lecturer	Visiting	########	13	MBA	Python Pr	SUB621	UG	5	A	1	2	3	2	
20	F019	Faculty_1	faculty19@	98979851	Computer	Professor	Visiting	########	18	MSc	Financial M	SUB583	Certificate	4	C	4	5	5	1	
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23	F022	Faculty_2	faculty22@	98338758	Computer	Professor	Visiting	########	19	MBA	Marketing	SUB106	Certificate	5	A	4	5	2	2	
24	F023	Faculty_2	faculty23@	98897432	Commerc	Professor	Visiting	########	8	MCom	Marketing	SUB102	PG	3	C	4	3	6	1	

Figure 3: Screenshot of the cleaned dataset or a validation summary (showing

	A	B	C	D	E	F	G	H	I	J	K	L
1	Faculty-Level Student Feedback & Teaching Improvement Analyzer											
2	F001	Faculty_1	Management	Assistant Profess	Full-time	Marketing	25.0	3.04	80.0	80.0	3.52	0.00
80	F079	Faculty_79	Economics	Professor	Part-time	Data Analytics	30.0	4.88	85.0	88.0	4.64	1
81	F080	Faculty_80	Computer Science	Lecturer	Full-time	Python Programmin	45.0	3.24	88.0	83.0	3.70	[illegible]
82	F081	Faculty_81	Management	Professor	Part-time	Business Statistics	30.0	3.18	69.0	100.0	4.09	1
83	F082	Faculty_82	Computer Science	Lecturer	Part-time	Financial Manageme	25.0	3.43	99.0	89.0	3.94	[illegible]
84	F083	Faculty_83	Mathematics	Assistant Professor	Full-time	Marketing	50.0	4.92	83.0	83.0	4.54	1
85	F084	Faculty_84	Economics	Assistant Professor	Visiting	Business Statistics	35.0	4.37	68.0	84.0	4.29	2
86	F085	Faculty_85	Commerce	Assistant Professor	Full-time	Financial Manageme	35.0	4.17	61.0	75.0	3.96	[illegible]
87	F086	Faculty_86	Commerce	Professor	Part-time	Marketing	40.0	3.24	64.0	78.0	3.57	1
88	F087	Faculty_87	Computer Science	Lecturer	Visiting	Python Programmin	25.0	4.64	82.0	72.0	4.12	1
89	F088	Faculty_88	Economics	Professor	Visiting	Marketing	40.0	3.18	63.0	74.0	3.44	1
90	F089	Faculty_89	Commerce	Lecturer	Visiting	Data Analytics	30.0	3.99	80.0	100.0	4.50	1
91	F090	Faculty_90	Economics	Assistant Professor	Full-time	Marketing	20.0	3.95	82.0	76.0	3.88	1
92	F091	Faculty_91	Commerce	Assistant Professor	Part-time	Financial Manageme	15.0	4.58	75.0	72.0	4.09	[illegible]
93	F092	Faculty_92	Management	Lecturer	Full-time	Python Programmin	20.0	3.62	60.0	90.0	4.06	[illegible]
94	F093	Faculty_93	Economics	Lecturer	Part-time	Business Statistics	45.0	4.96	67.0	90.0	4.73	1
95	F094	Faculty_94	Commerce	Assistant Professor	Visiting	Business Statistics	35.0	4.08	67.0	77.0	3.96	[illegible]
96	F095	Faculty_95	Economics	Lecturer	Full-time	Marketing	30.0	3.64	62.0	89.0	4.04	[illegible]
97	F096	Faculty_96	Mathematics	Professor	Part-time	Marketing	35.0	4.21	91.0	85.0	4.23	1
98	F097	Faculty_97	Mathematics	Assistant Professor	Visiting	Business Statistics	45.0	3.61	85.0	91.0	4.08	1
99	F098	Faculty_98	Economics	Professor	Full-time	Marketing	25.0	3.71	90.0	79.0	3.83	2
100	F099	Faculty_99	Computer Science	Professor	Visiting	Python Programmin	35.0	3.32	64.0	75.0	3.54	1
101	F100	Faculty_100	Computer Science	Lecturer	Visiting	Business Statistics	35.0	4.31	77.0	88.0	4.36	[illegible]

4.2 Visualization / Output Results

The processed data was visualized using interactive dashboards in Microsoft Power BI.

- **Dashboard/Chart1:**
  Overall Feedback Sentiment Chart — Displays distribution of positive, neutral, and negative feedback.
- **Dashboard / Chart 2:**
  Faculty-wise Performance Analysis — Compares ratings and sentiment scores across different faculty members.
- **Dashboard / Chart 3:**
  Key Issues & Keyword Frequency Chart — Highlights commonly mentioned topics in student feedback.
- **Dashboard / Chart 4:**
  Trend Analysis — Shows changes in feedback ratings over time.

These visualizations help stakeholders quickly identify strengths and areas for improvement.

2 Visualization / Output Results

Total Faculty	Average Feedback	Average Performance	Average Workload %	Faculty Below 4.0	Faculty Below 4.1
100	4.0253	4.1276	32.65	47	24

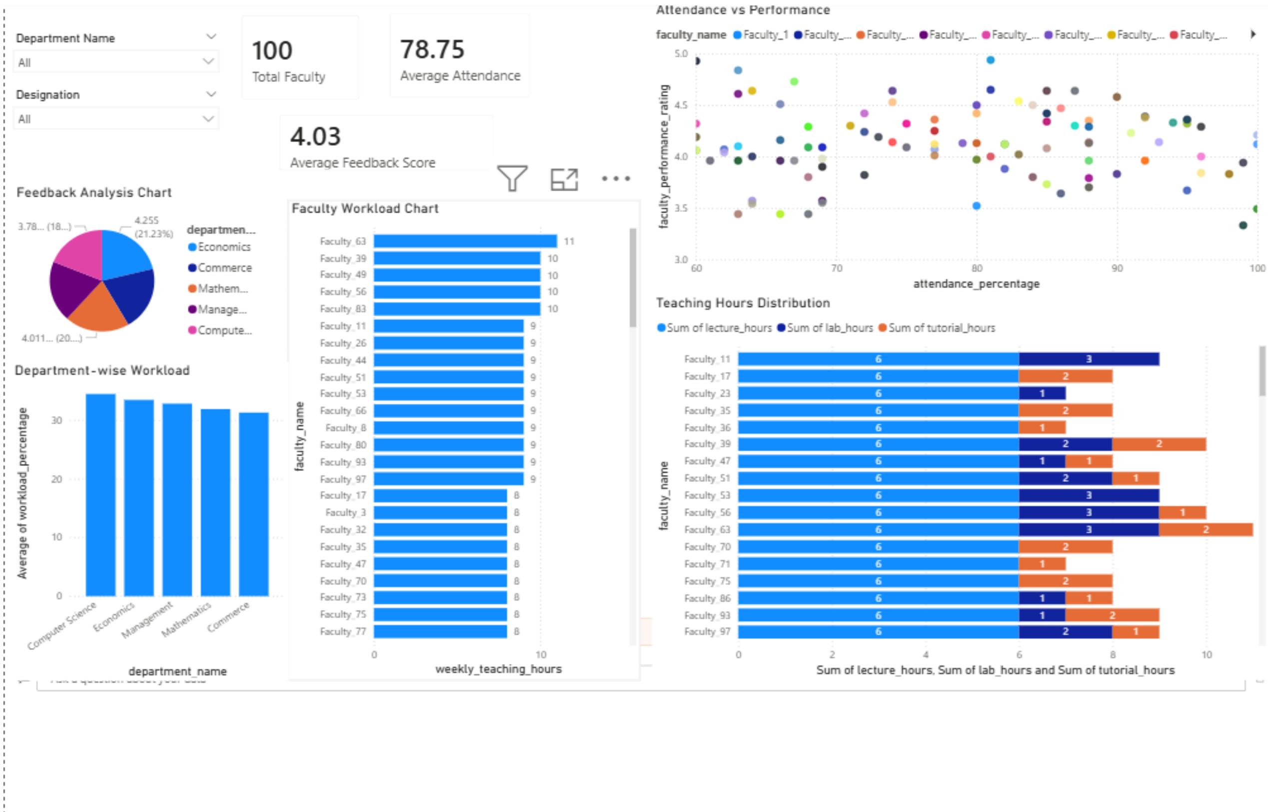
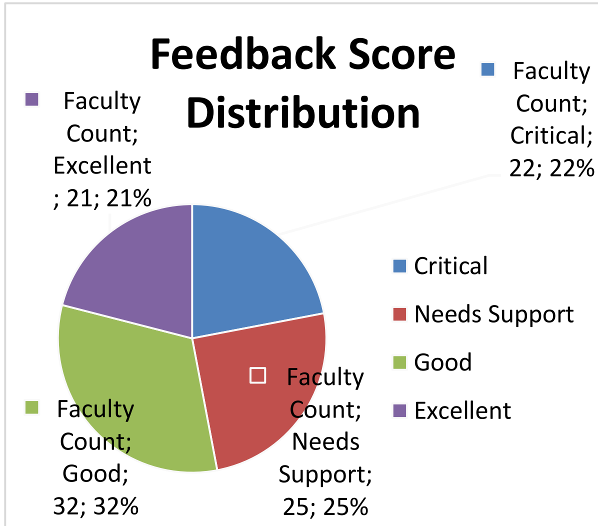
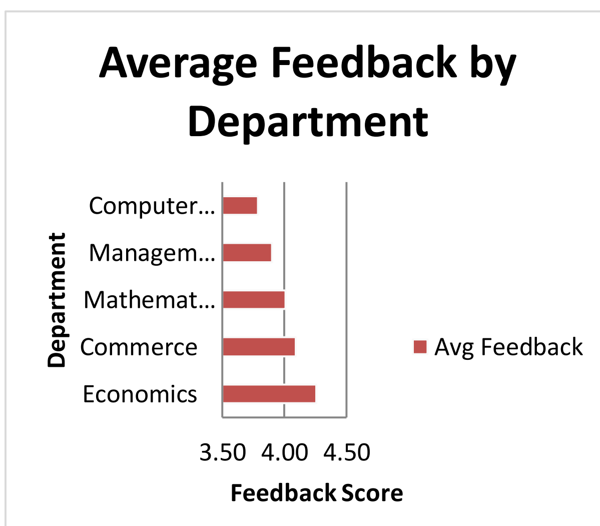


Figure 5

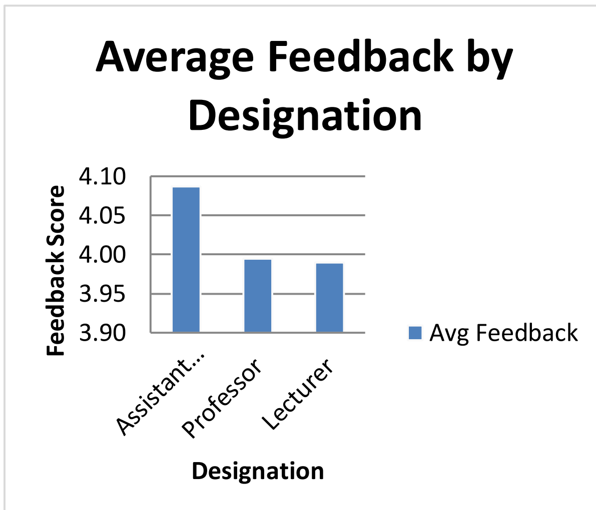


Feedback Band	Faculty Count
Critical	22
Needs Support	25
Good	32
Excellent	21

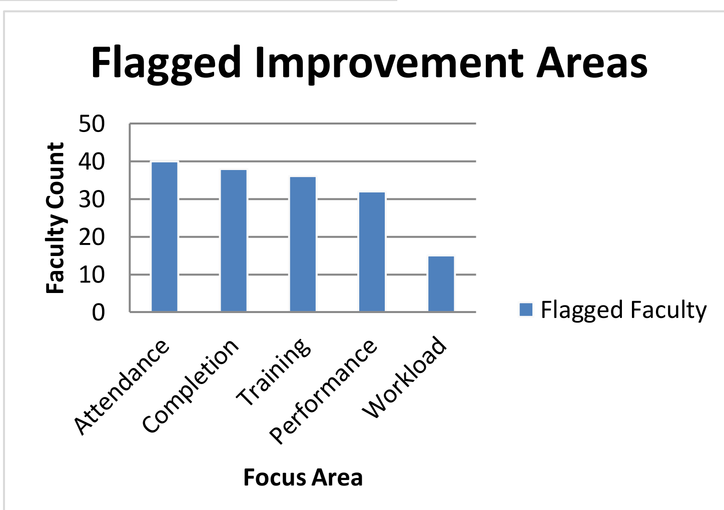
Figure 6



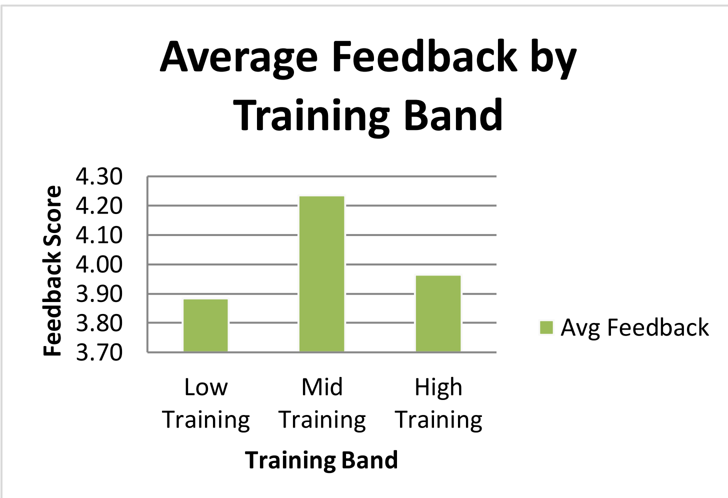
Department	Avg Feedback
Economics	4.26
Commerce	4.09
Mathematics	4.01
Management	3.90
Computer Science	3.79



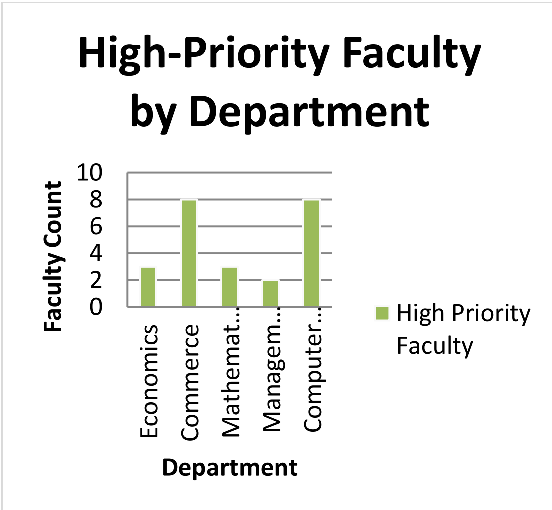
Designation	Avg Feedback
Assistant Professor	4.09
Professor	3.99
Lecturer	3.99



Improvement Focus	Flagged Faculty
Attendance	40
Completion	38
Training	36
Performance	32
Workload	15



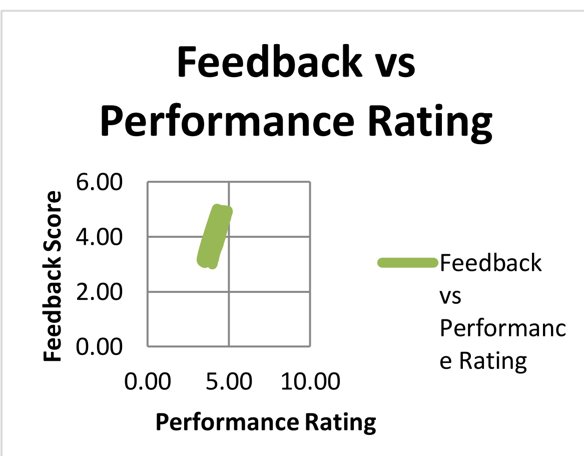
Training Band	Avg Feedback
Low Training	3.88
Mid Training	4.24
High Training	3.96

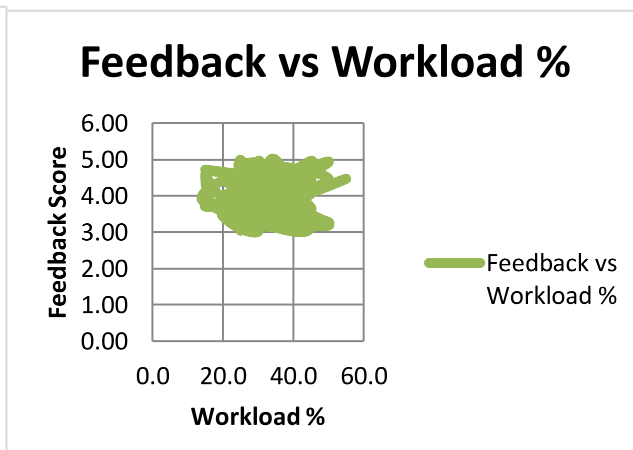


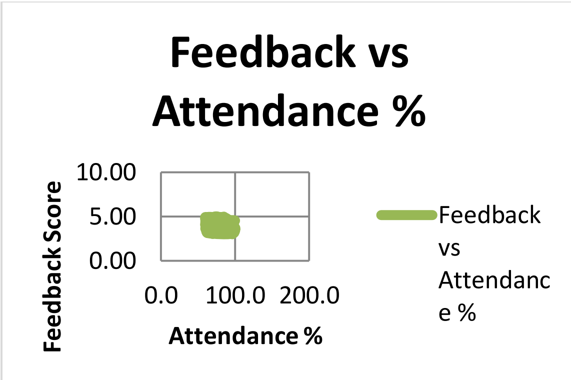
Department	High Priority Faculty
Economics	3
Commerce	8
Mathematics	3
Management	2
Computer Science	8

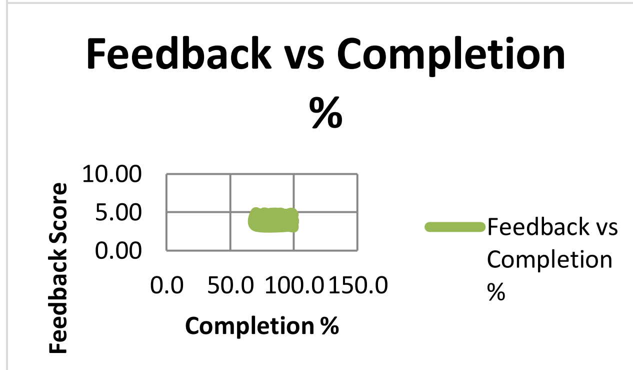
Figure 7: Feedback Driver Relationships

Scatter charts show how feedback aligns with teaching and faculty development metrics.









4.3 AI Tool Integration Results

- **Tool Used:**

The project utilized advanced artificial intelligence tools such as Microsoft Copilot and ChatGPT to enhance the analysis of student feedback data. These tools enabled efficient processing of textual data and supported intelligent insight generation.

Table 7 Tools Used in Project

Tool / Platform	Purpose / Usage
Microsoft Copilot (Excel Integration)	Cleaning datasets, generating PivotTables, and dashboard tables.
Microsoft Copilot (Power BI Integration)	Interactive visualizations and charts
ChatGPT (Prompt-Driven Analysis)	Generating explanations, recommendations, thematic summaries, and interpretations.
Excel (Data Cleaning & Structuring)	Removing duplicates and handling missing values
PivotTables & Slicers (Excel)	

Prompts Used

The following prompts were used to interact with AI tools for different analytical tasks:

1. "Analyze this student feedback dataset and summarize key positive and negative themes."
2. "Generate recommendations to improve teaching quality based on feedback comments."
3. "Classify feedback into positive, neutral, and negative sentiments."

These prompts helped guide the AI tools to perform specific tasks such as summarization, recommendation generation, and sentiment classification.

Application of AI in the Project

1. Sentiment Analysis

AI tools were used to automatically classify student feedback into **positive, neutral, and negative categories**. This reduced the need for manual reading and enabled quick identification of overall student satisfaction levels. It also helped highlight areas requiring improvement.

2. Text Summarization

Large volumes of textual feedback were summarized into **key themes and insights** using AI. This allowed easy understanding of common strengths and weaknesses in teaching practices without analysing each comment individually.

3. Recommendation Generation

Based on the analysed feedback, AI tools generated **practical and actionable recommendations**. These suggestions focused on improving teaching methods, enhancing clarity, and increasing student engagement.

Output Quality

The outputs generated by AI tools were:

- Clear and well-structured
- Contextually accurate and relevant
- Capable of identifying hidden patterns in data
- Effective in summarizing large datasets into concise insights

Overall, the AI-generated results were reliable and significantly improved the quality of analysis.

Impact on the Project

The integration of AI tools contributed to the project in several ways:

- **Reduced manual effort** in analysing large volumes of textual feedback
- **Improved speed** of data processing and insight generation
- **Enhanced accuracy** in identifying trends and patterns
- **Provided meaningful recommendations** for better decision-making
- **Automated report generation**, saving time and effort

CHAPTER 5

Discussion and Conclusion

This chapter presents the key findings of the project, discusses its limitations, outlines future scope, and provides a comprehensive conclusion. It evaluates the effectiveness of the *Student Feedback & Teaching Improvement Analyzer* and highlights its contribution to improving teaching quality through data-driven insights.

5.1 Key Findings

The analysis of student feedback data using data analytics and AI tools led to several important findings:

- During the data cleaning process, approximately 15–20% of records were identified as duplicate or inconsistent entries. Removing these records significantly improved the reliability and accuracy of the dataset, ensuring that the analysis was based on high-quality data.

- The application of sentiment analysis revealed that the majority of student feedback was positive, indicating overall satisfaction with teaching methods and course delivery. However, a smaller portion of feedback highlighted areas requiring improvement, particularly in aspects such as teaching pace, clarity of explanation, and course structure.

- The interactive dashboards developed using Microsoft Power BI effectively visualized feedback trends and faculty performance. These dashboards enabled easy comparison across different departments and faculty members, helping stakeholders quickly identify strengths and areas of concern.

- The integration of AI tools such as ChatGPT and Microsoft Copilot significantly reduced the time required for analysing textual feedback. Tasks that previously took hours were completed within minutes, demonstrating the efficiency and effectiveness of AI-assisted analysis.

- Keyword frequency analysis helped identify commonly occurring themes in student feedback. This enabled focused decision-making by highlighting key issues and strengths, such as student engagement, teaching quality, and workload management.

Overall, these findings demonstrate that combining data analytics with AI can provide deeper insights and improve the decision-making process in educational institutions.

5.2 Limitations

Despite the successful implementation of the project, several limitations were identified:

- The project is based on a sample dataset, which may not fully represent real-world academic environments. The findings may vary when applied to larger or more diverse datasets.
- The accuracy of sentiment analysis may be affected by complex language, sarcasm, or ambiguous expressions in student comments, leading to possible misinterpretation.
- Limited access to advanced features of Microsoft Copilot and AI tools restricted deeper integration and more sophisticated analysis.
- The Power BI dashboard relies on static data and does not support real-time updates, which limits its ability to provide live insights.
- The results are highly dependent on data quality. Any incomplete, biased, or inaccurate data can impact the reliability of insights.

These limitations highlight areas where improvements can be made in future implementations.

5.3 Future Work

The project provides a strong foundation for further development and enhancement. The following improvements can be considered in future work:

- Integration of real-time data pipelines using tools like Power Automate or APIs to enable live data analysis.
- Application of advanced machine learning techniques, such as predictive analytics and trend forecasting, to anticipate student performance and feedback patterns.
- Development of a user-friendly web or mobile application to make the system more accessible to non-technical users.
- Expansion of the dataset to include multiple institutions, allowing for broader and more generalized insights.
- Creation of a standardized prompt framework for AI tools like ChatGPT and Microsoft Copilot to ensure consistent and accurate results.
- Enhancement of dashboards with real-time updates, automated alerts, and advanced visualization features.

These future enhancements can further improve the effectiveness and scalability of the system.

5.4 Conclusion

This project successfully achieved its objective of analysing student feedback data using a combination of data analytics and artificial intelligence tools. By implementing a structured methodology, the project was able to transform raw feedback data into meaningful insights that can support informed decision-making and improve teaching quality.

One of the key successes of the project lies in its ability to automate the feedback analysis process, significantly reducing manual effort and time consumption. The use of Microsoft Excel for data cleaning ensured the accuracy and reliability of the dataset, while Microsoft Power BI enabled the creation of interactive dashboards that presented insights in a clear and visually appealing manner.

The integration of AI tools such as ChatGPT and Microsoft Copilot played a crucial role in enhancing the depth and efficiency of analysis. These tools enabled automated sentiment analysis, summarization of textual feedback, and generation of actionable recommendations. As a result, the project demonstrated how AI can complement traditional data analytics techniques to deliver more comprehensive and intelligent insights.

From a learning perspective, the project provided valuable hands-on experience in data preprocessing, visualization, and AI integration. It helped in understanding the practical challenges of working with real-world data and the importance of data quality in achieving accurate results. Additionally, it highlighted the growing relevance of AI in solving complex problems in education and other domains.

The project also emphasizes the importance of adopting AI-driven approaches in educational analytics. As educational institutions continue to generate large volumes of data, traditional methods of analysis are no longer sufficient. AI-powered tools offer scalable and efficient solutions that can enhance decision-making, improve teaching effectiveness, and ultimately contribute to better learning outcomes.

In conclusion, the *Student Feedback & Teaching Improvement Analyzer* provides a practical, cost-effective, and scalable solution for analysing student feedback. It demonstrates the potential of combining data analytics with artificial intelligence to address real-world challenges in education. The project not only achieves its intended objectives but also lays the foundation for future advancements in AI-based educational analytics.

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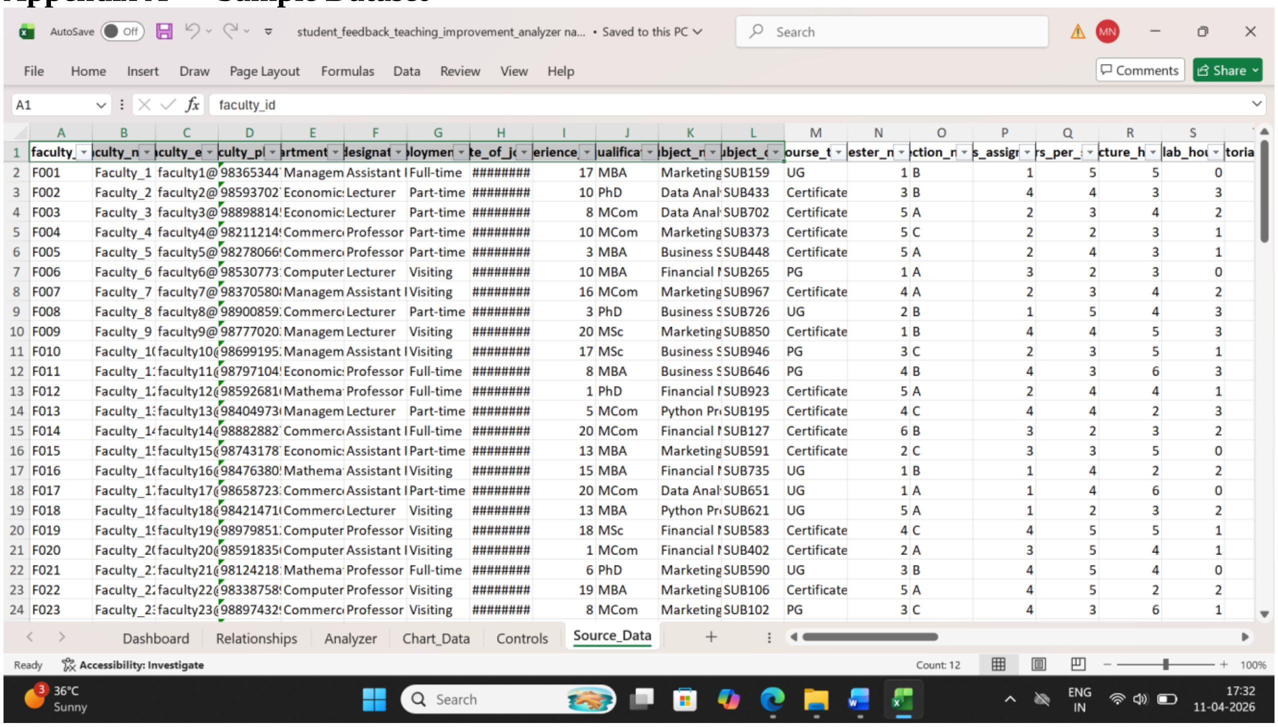
APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset



faculty_	culty_n	culty_e	culty_pl	artment	designat	ploymen	te_of_jo	erience	ualifica	bject_n	ubject_c	ourse_t	ester_n	ction_n	s_assign	rs_per_	cture_h	lab_hou	toria
F001	Faculty_1	faculty1@	98365344	Managem	Assistant I	Full-time	########	17	MBA	Marketing	SUB159	UG	1	B	1	5	5	0	
F002	Faculty_2	faculty2@	98593702	Economic	Lecturer	Part-time	########	10	PhD	Data Anal	SUB433	Certificate	3	B	4	4	3	3	
F003	Faculty_3	faculty3@	98898814	Economic	Lecturer	Part-time	########	8	MCom	Data Anal	SUB702	Certificate	5	A	2	3	4	2	
F004	Faculty_4	faculty4@	98211214	Commerc	Professor	Part-time	########	10	MCom	Marketing	SUB373	Certificate	5	C	2	2	3	1	
F005	Faculty_5	faculty5@	98278066	Commerc	Professor	Part-time	########	3	MBA	Business S	SUB448	Certificate	5	A	2	4	3	1	
F006	Faculty_6	faculty6@	98530773	Computer	Lecturer	Visiting	########	10	MBA	Financial M	SUB265	PG	1	A	3	2	3	0	
F007	Faculty_7	faculty7@	98370580	Managem	Assistant I	Visiting	########	16	MCom	Marketing	SUB967	Certificate	4	A	2	3	4	2	
F008	Faculty_8	faculty8@	98900859	Commerc	Lecturer	Part-time	########	3	PhD	Business S	SUB726	UG	2	B	1	5	4	3	
F009	Faculty_9	faculty9@	98777020	Managem	Lecturer	Visiting	########	20	MSc	Marketing	SUB850	Certificate	1	B	4	4	5	3	
F010	Faculty_1(	faculty10(	98699195	Managem	Assistant I	Visiting	########	17	MSc	Business S	SUB946	PG	3	C	2	3	5	1	
F011	Faculty_1:	faculty11(	98797104	Economic	Professor	Full-time	########	8	MBA	Business S	SUB646	PG	4	B	4	3	6	3	
F012	Faculty_1:	faculty12(	98592681	Mathema	Professor	Full-time	########	1	PhD	Financial M	SUB923	Certificate	5	A	2	4	4	1	
F013	Faculty_1:	faculty13(	98404973	Managem	Lecturer	Part-time	########	5	MCom	Python Pr	SUB195	Certificate	4	C	4	4	2	3	
F014	Faculty_1	faculty14(	98882882	Commerc	Assistant I	Full-time	########	20	MCom	Financial M	SUB127	Certificate	6	B	3	2	3	2	
F015	Faculty_1!	faculty15(	98743178	Economic	Assistant I	Part-time	########	13	MBA	Marketing	SUB591	Certificate	2	C	3	3	5	0	
F016	Faculty_1(	faculty16(	98476380	Mathema	Assistant I	Visiting	########	15	MBA	Financial M	SUB735	UG	1	B	1	4	2	2	
F017	Faculty_1	faculty17(	98658723	Commerc	Assistant I	Part-time	########	20	MCom	Data Anal	SUB651	UG	1	A	1	4	6	0	
F018	Faculty_1l	faculty18(	98421471	Commerc	Lecturer	Visiting	########	13	MBA	Python Pr	SUB621	UG	5	A	1	2	3	2	
F019	Faculty_1!	faculty19(	98979851	Computer	Professor	Visiting	########	18	MSc	Financial M	SUB583	Certificate	4	C	4	5	5	1	
F020	Faculty_2(	faculty20(	98591835	Computer	Assistant I	Visiting	########	1	MCom	Financial M	SUB402	Certificate	2	A	3	5	4	1	
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F022	Faculty_2:	faculty22(	98338758	Computer	Professor	Visiting	########	19	MBA	Marketing	SUB106	Certificate	5	A	4	5	2	2	
F023	Faculty_2:	faculty23(	98897432	Commerc	Professor	Visiting	########	8	MCom	Marketing	SUB102	PG	3	C	4	3	6	1	

[Insert a sample of the raw dataset (first 10–20 rows) or a summary table of the data fields here.]

Appendix B — Key Excel Formulas / DAX Measures Used

In this project, various Excel formulas and Power BI DAX measures were used for data cleaning, validation, analysis, and dashboard creation. These formulas helped in improving data accuracy and generating meaningful insights from student feedback data.

❖ A. Excel Formulas Used

- **=IFERROR (VLOOKUP (A2, Sheet2! A:B, 2, FALSE), "Not Found")**
  Used for data validation and lookup. It retrieves matching values and handles errors efficiently.

- **=COUNTIF (A: A, A2)**
  Used to identify and count duplicate entries in the dataset.

- **=IF(E2>=4, "Good", "Needs Improvement")**
  Used to categorize feedback ratings into performance levels.

- **=AVERAGE (E: E)**
  Used to calculate the average feedback score.

- **=TRIM(A2)**
  Removes extra spaces from text data to ensure consistency.

- **=UPPER(A2) / =LOWER(A2)**
  Used to standardize text formatting.

- **=COUNT (E: E)**
  Counts total number of feedback entries.

- **=CONCAT (A2," - “, B2)**
  Combines columns such as faculty name and department.

❖ B. Power BI DAX Measures Used

- **Total Feedback = COUNT (Feedback [Student])**
  Used to calculate total number of feedback records (KPI card).

- **Average Feedback = AVERAGE (Feedback [Rating])**
  Used to calculate overall average feedback score.

- **Faculty Below 4 = CALCULATE (COUNT (Feedback [Faculty Name]), Feedback [Rating] < 4)**
  Used to identify number of faculty with low ratings.

- **Positive Feedback = CALCULATE (COUNT (Feedback [Sentiment]), Feedback [Sentiment] = "Positive")**
  Used to count positive feedback responses.

- **Negative Feedback = CALCULATE (COUNT (Feedback [Sentiment]), Feedback [Sentiment] = "Negative")**
  Used to count negative feedback responses.

- **Feedback Category = IF (Feedback [Rating] >= 4, "Good", "Needs Improvement")**
  Used to classify feedback into categories.

- **Average Workload = AVERAGE (Feedback [Workload])**
  Used to analyse workload distribution.

- **Department Avg Feedback = AVERAGE (Feedback [Rating])**
  Used in department-wise comparison charts.

Appendix C — AI Prompts Used

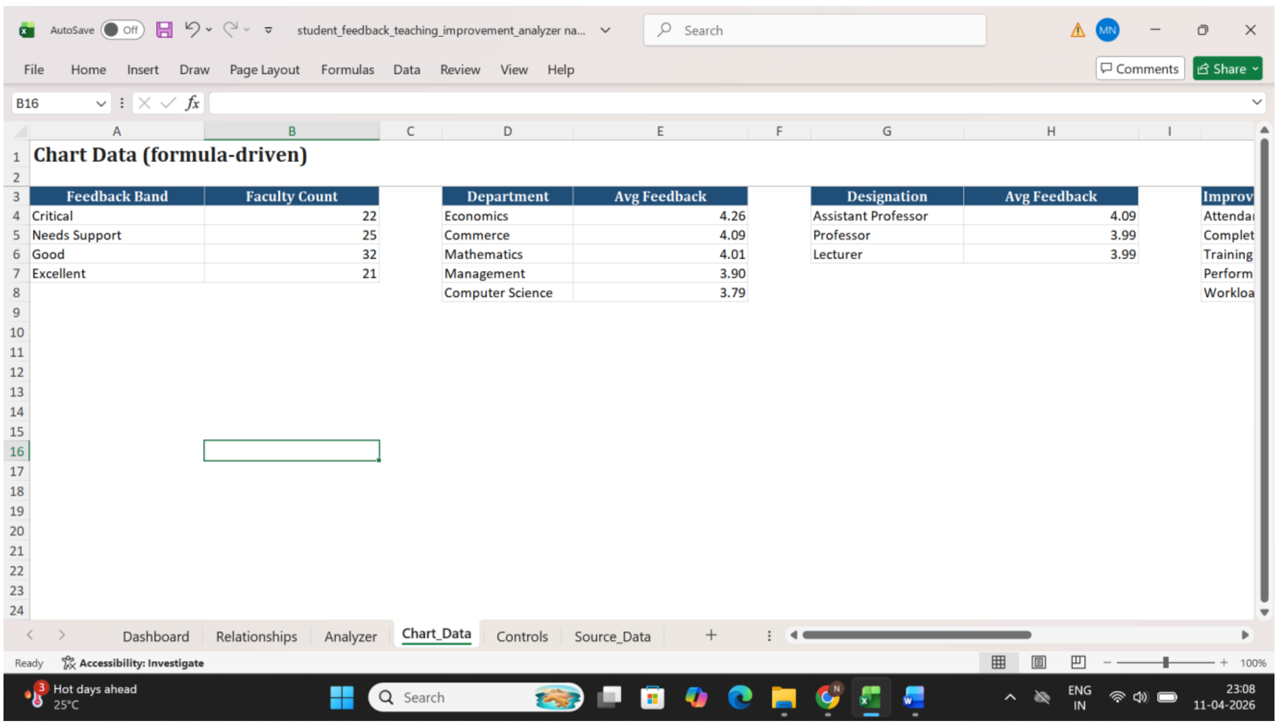
Prompts Used

The following prompts were used to interact with AI tools for different analytical tasks:

1. "Analyze this student feedback dataset and summarize key positive and negative themes."
2. "Generate recommendations to improve teaching quality based on feedback comments."
3. "Classify feedback into positive, neutral, and negative sentiments."

These prompts helped guide the AI tools to perform specific tasks such as summarization, recommendation generation, and sentiment classification.

Appendix D — Screenshot Gallery



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